SOFTWARE ARCHITECTURE & DESIGN SPECIFICATION

Railway Reservation System

Team 12

SRN	Name
PES2UG24CS345	PAVAN KUMAR H
PES2UG24CS352	POOJA KOPPAD
PES2UG24CS351	POOJA KANAKAPPANAVARA
PES2UG24CS354	POTHINENI DINESH

Revision History

Version	Date	Description
1.0	September 2026	Initial Version

1.Introduction

1.1 Purpose

This document is a high-level architectural and design overview of Railway Reservation System. It describes the principal design elements, their roles and interactions, security architecture, interfaces, and fault handling and is traceable to the software requirements as captured in Software Requirements Specification (SRS).

1.2 Scope

We will discuss a Railway reservation system as application on top of C++17.

The system supports:

• User registration and login

• Train search

• Seat availability checking

• Passenger details entry

• Fare calculation

• Ticket booking

• PNR generation and enquiry

• Booking history

• Ticket cancellation

• Refund processing

• Waitlist handling

• Administrative train and fare management

• Reservation and revenue reports

It utilizes local persistent storage. The system does not include the actual operations of a railway system, nor any settlement of real external payments.

1.3 Intended Audience

This document is intended for:

• Developers

• Testers

• Project team members

• Project guide

• Evaluators

Architectural Overview

2.1 Architecture Pattern

It employs a Layered Architecture system.

The major layers are:

• Presentation Layer

• Application Layer

• Business Logic Layer

• Persistence Layer

User interaction is handled by the Presentation Layer. Reservation operations are processed under Application and Business Logic layers. The Persistence Layer is used to manage local data storage.

2.2 Architectural Goals

The architecture is designed to:

• Separate different system responsibilities.

• Maintain correct seat availability.

• Support booking and cancellation operations.

• Protect user and reservation information.

• Keep simple and clear interfaces between components.

• Individual components that are easier to test and maintain

2.3 Architectural Constraints

• Our system is implemented in C++17.

• The application is console based.

• It persists data with local file storage.

• Payment processing is simulated.

• Network communication is not required.

• Authorized users have access to administrative operations.

System Architecture

3.1 System Components

The key components of Railway Reservation System Component are:

• Console UI / Main

• AuthManager

• TrainSearch

• SeatInventory

• BookingManager

• PaymentModule

• CancellationManager

• PNRService / HistoryService

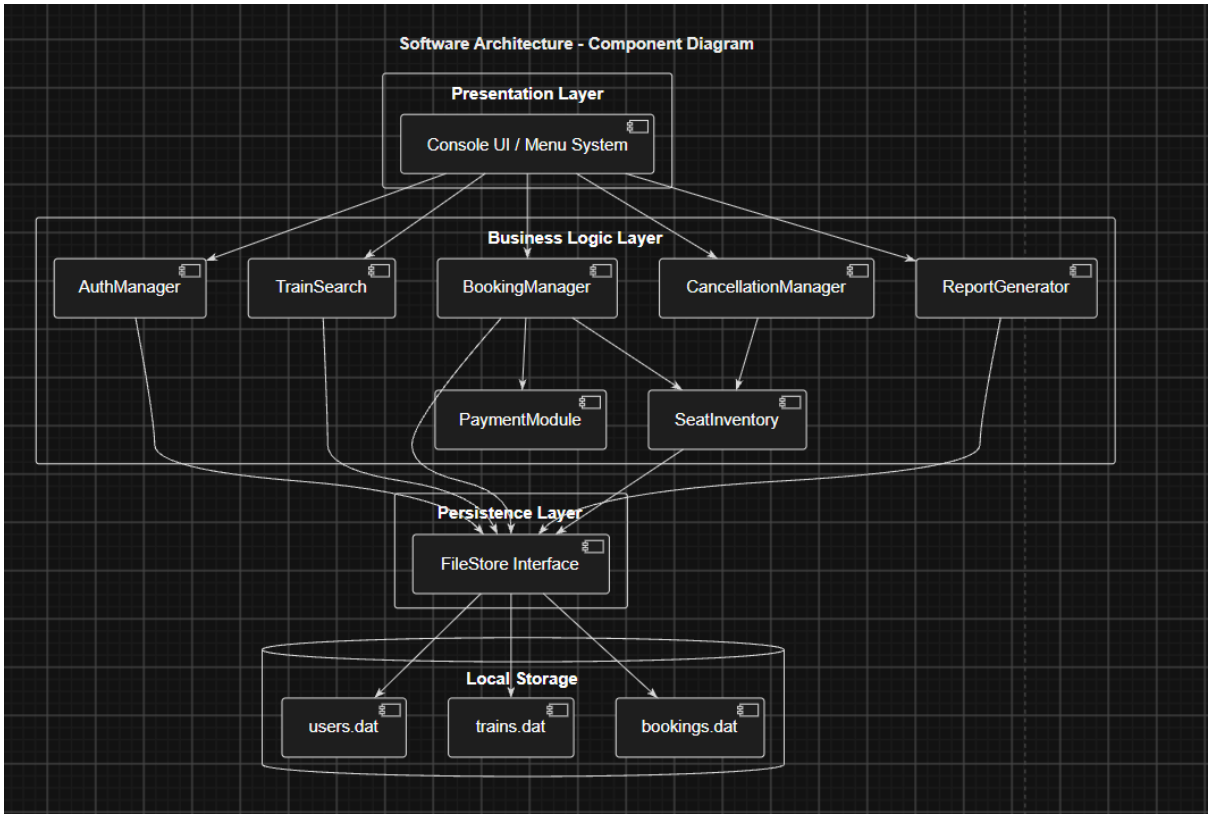
• TicketPrinter

• AdminTrainManager / AdminFareManager

• ReportGenerator

• FileStore

3.2 Component Diagram



Component Description

Component Responsibility

Console UI / Main receives input from the user and displays output from the system

AuthManager Handles registration, login and authentication

TrainSearch Searches and shows trains

SeatInventory Perform checks, hold, release and update actions on seat availability

BookingManager Coordinates ticket booking

PaymentModule Responsible for fare calculation, simulated payment and refund calculation

CancellationManager To manage ticket cancellation and related refund processing

PNRService / HistoryService Fetches PNR, booking status and booking history

TicketPrinter Generates ticket/e-ticket information

AdminTrainManager/AdminFareManager Manages train and fare information

ReportGenerator Generates reservation and report information

FileStore Manages persistent storage and localization of records.

4.Architecture Traceability

Requirement Area	Design Component
User registration and login	AuthManager
Train search	TrainSearch
Seat availability	SeatInventory
Passenger details	BookingManager
Fare calculation	PaymentModule
Ticket booking	BookingManager
PNR generation and enquiry	PNRService
Booking history	HistoryService
Ticket cancellation	CancellationManager
Refund processing	PaymentModule / CancellationManager
Train management	AdminTrainManager
Fare and quota management	AdminFareManager
Reports	ReportGenerator
Persistent storage	FileStore
Authentication security	AuthManager
Input validation	Console UI / Application Components
Reservation data integrity	BookingManager / SeatInventory / FileStore

---

5. Security Architecture

5.1 Security Objectives

The system security design aims to:

1. Protect user credentials and reservation information from unauthorized access.
2. Maintain the integrity and confidentiality of reservation and cancellation information.
3. Restrict administrative operations to authorized users.

5.2 Authentication and Authorization

AuthManager handles user authentication.

Users must successfully authenticate before accessing protected reservation information.

Administrative functions are accessible only to authorized administrator accounts.

5.3 Password Protection

Passwords must not be stored or displayed in plaintext.

AuthManager is responsible for protecting credential information before it is stored.

5.4 Input Validation

User input is validated before processing or storage.

Validation is applied to:

- Login information
- Passenger information
- Train search information
- Booking information
- Administrative input

Invalid input should result in an appropriate error message.

5.5 Data Protection and Integrity

FileStore manages access to persistent files.

The system must maintain consistency between booking information and seat availability.

A failed booking or payment operation must not leave an incorrect confirmed reservation.

A successful cancellation must update the booking status and corresponding seat availability.

6. Detailed Design

6.1 AuthManager

Responsibilities:

- Perform user registration
- Perform user authentication
- Perform user login/logout management
- Perform session validation
- Support admin authentication

6.2 TrainSearch

Responsibilities:

- Perform train searching

- Perform source and destination validation
- Perform journey date validation
- Display train information
- Display available classes and rates

6.3 SeatInventory

Responsibilities:

- Perform seat status checking
- Perform seat holding
- Perform seat cancellations
- Perform updates on seat status

6.4 BookingManager

Responsibilities:

- Perform reservation creation
- Verify booking information
- Coordinate seat assignments
- Coordinate payments
- Confirm booking completion

6.5 PaymentModule

Responsibilities:

- Calculate costs
- Conduct payment processing
- Calculate refunds

6.6 CancellationManager

Responsibilities:

- Validate cancellations
- Cancel reservations
- Calculate refunds
- Free up seats
- Update reservation status

6.7 PNRService / HistoryService

Responsibilities:

- Generate PNR
- Retrieve reservation using PNR
- Display reservation status
- Display reservation history

6.8 TicketPrinter

Responsibilities:

- Purpose Spot ticket generation
- Display PNR, train, passenger and price information
- Provide e-ticket printing

6.9 AdminTrainManager / AdminFareManager

Responsibilities:

- Update train information
- Manage train schedules
- Manage fare settings
- Manage quotas
- Manage train status

6.10 ReportGenerator

Responsibilities:

- Prepare occupancy reports
- Prepare financial reports

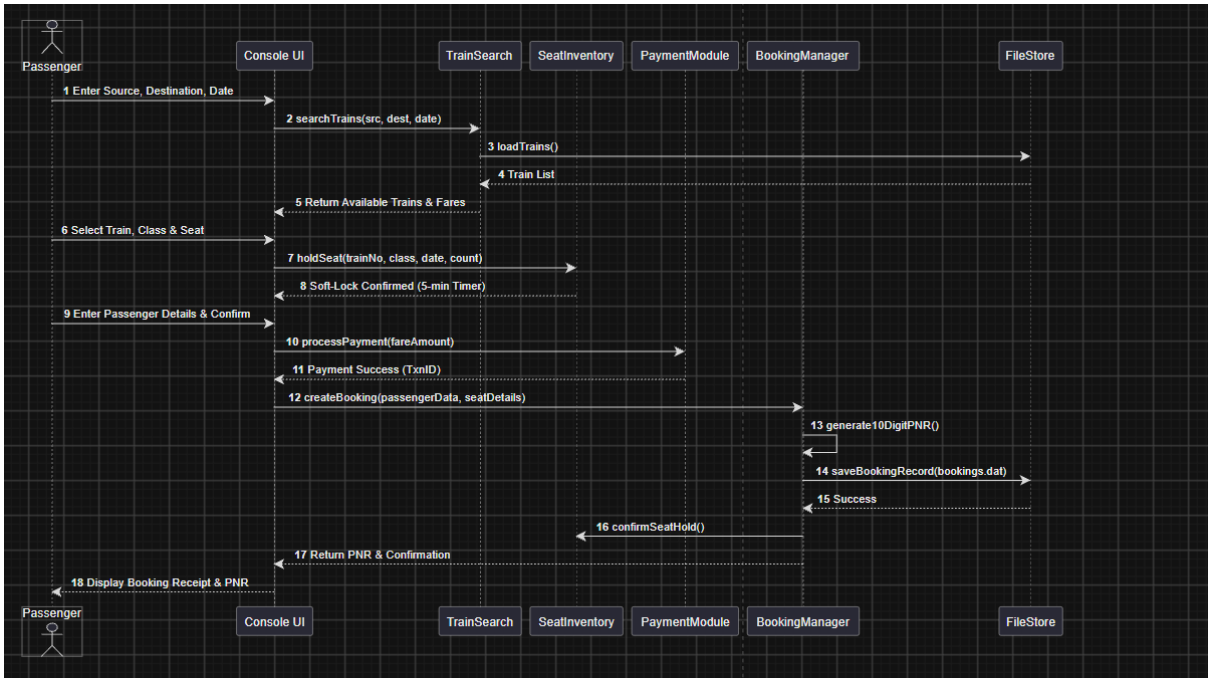
6.11 FileStore

Responsibilities:

- Read records
- Write records
- Maintain user information
- Maintain train information
- Maintain reservation information

7. UML Sequence Diagrams

7.1 Ticket Booking and Payment



The ticket booking sequence includes the following primary steps:

1. The passenger performs a train search.

2. TrainSearch fetches train information matching the search.

3. The passenger picks a particular train and class.

4. SeatInventory checks to see if there are seats available.

5. The Booking manager collects information about the passenger.

6. The Payment Module computes the fare.

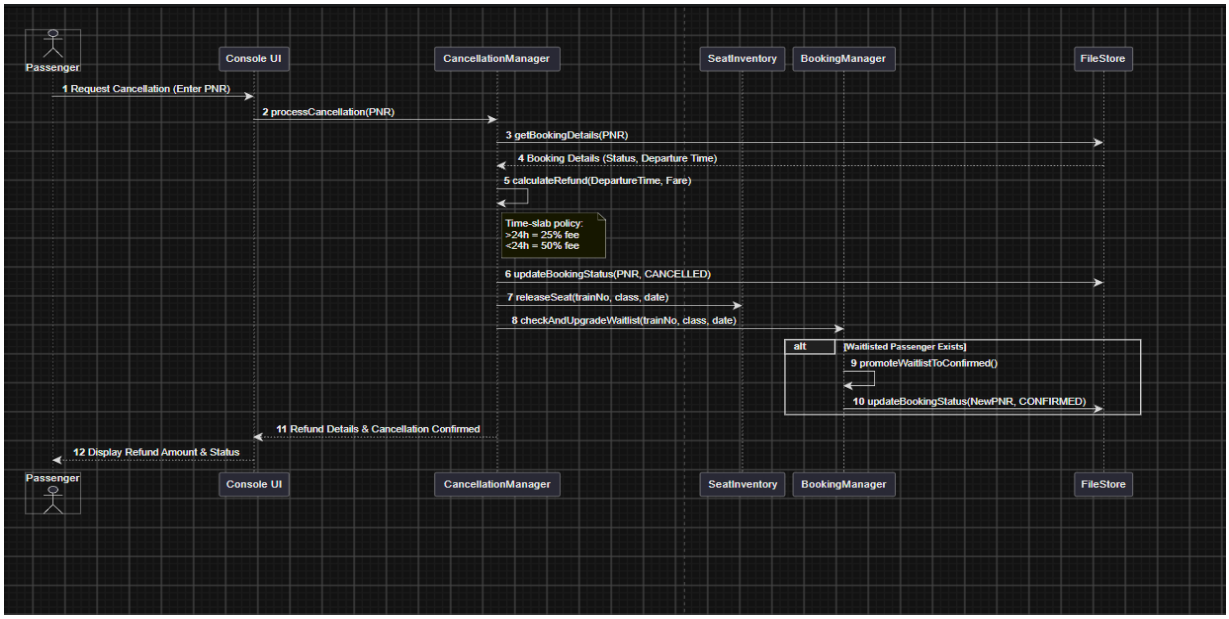
7. The payment is completed.

8. PNRService produces PNR.

9. The Booking manager saves the confirmed booking in the FileStore.

10. The passenger gets the confirmation along with the ticket details.

7.2 Ticket Cancellation and Refund



The cancellation process contains the following key interactions:

1. The passenger inputs the PNR.

2. The PNRService obtains the booking details.

3. The passenger approves the cancellation.

4. CancellationManager authenticates the reservation.

5. The Payment Module figures out the refund due.

6. SeatInventory makes the seat available again.

7. The booking information is modified using FileStore.

8. In case of waitlisted reservations, the necessary actions are taken.

9. The cancellation/refund information is communicated to the passenger.

8. API and Interface Design

Component	Interface	Purpose	
AuthManager	registerUser()	Register user	
AuthManager	login()	Authenticate user	
AuthManager	logout()	End session	
TrainSearch	searchTrains()	Search trains	
TrainSearch	getTrainDetails()	Retrieve train details	
SeatInventory	checkAvailability()	Check available seats	
SeatInventory	holdSeat()	Temporarily hold seat	
SeatInventory	releaseSeat()	Release seat	
BookingManager	createBooking()	Create reservation	
BookingManager	confirmBooking()	Confirm reservation	
PaymentModule	calculateFare()	Calculate fare	

PaymentModule	processPayment()	Process simulated payment	
PaymentModule	calculateRefund()	Calculate refund	
CancellationManager	cancelBooking()	Cancel booking	
PNRService	generatePNR()	Generate PNR	
PNRService	getBookingByPNR()	Retrieve booking	
FileStore	readData()	Read persistent data	
FileStore	writeData()	Write persistent data	

---

9. Error Handling

The system performs error handling of invalid operations through validations and by showing the appropriate error messages.

Authentication Errors

- Invalid username and password
- Duplicate registration
- Unauthorized access to administrator

Search Errors

- Invalid search specifications
- Invalid date
- There are no matching trains

Booking Errors

- No seats available
- Invalid passenger specifications
- Payment failure
- The seat hold has expired

Cancellation Errors

- Invalid PNR
- Booking does not exist
- Invalid cancellation request

File Errors

- File does not exist
- Invalid record
- File reading/writing failure
- Data corruption

The system should be able to handle these errors without crashing.

10. Design Traceability Matrix

REQUIREMENT	DESIGN COMPONENT
REGISTRATION	AUTHMANAGER
LOGIN	AUTHMANAGER
TRAIN SEARCH	TRAINSEARCH
SEAT AVAILABILITY	SEATINVENTORY
BOOKING	BOOKINGMANAGER
PAYMENT	PAYMENTMODULE
PNR	PNRSERVICE
BOOKING HISTORY	HISTORYSERVICE
CANCELLATION	CANCELLATIONMANAGER
REFUND	PAYMENTMODULE
TRAIN MANAGEMENT	ADMINTRAINMANAGER
FARE MANAGEMENT	ADMINFAREMANAGER
REPORTS	REPORTGENERATOR
DATA PERSISTENCE	FILESTORE
AUTHENTICATION SECURITY	AUTHMANAGER
INPUT VALIDATION	APPLICATION COMPONENTS
DATA INTEGRITY	BOOKINGMANAGER, SEATINVENTORY, FILESTORE

---

11. Conclusion

The Railway Reservation System comprises a layered architecture including presentation, application, business logic and persistence.

The core components separately deal with authentication, train search, seat inventory, ticketing, payments, cancellations, PNR/history, administration, reporting and data storage.

The design satisfies the functional, non-functional and safety requirements specified in the SRS while providing clear interactions, internal interfaces, security controls and error handling.